Water Quality Prediction in Aquaculture Using Machine Learning and Internet Of Things

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Abstract. The expansion of Internet of Things (IoT) technology has opened new avenues for monitoring and managing various aspects of aquaculture. The IoT-based water monitoring system integrates sensors to collect real-time data on water quality parameters such as temperature, pH levels, dissolved oxygen, turbidity crucial for aquatic life. Through a wireless network, the collected data is transmitted to a central server for analysis and interpretation. Leveraging cloud computing and data analytics, the WQPA provides actionable insights to aqua culturists, enabling proactive decision-making and precise adjustments to optimize water pollution due to side effects of several human activities. This enhances the overall health and growth of aquatic organisms, leading to improved productivity and sustainability in aquaculture operations. The IoT-based WQPA offers a scalable, cost-effective solution to address the evolving challenges in aquaculture management, promoting efficient resource utilization and environmental conservation. WQPA works on sustainable development goals 8 and 14 where the life below water is protected to enhance the economic growth of fish farmers.

Keywords: Decision Support Systems, Machine Learning, Sustainable Management, Aquaculture, and Water Quality Prediction, Dissolved Oxygen

1. INTRODUCTION

To meet the growing demand for seafood and lessen the strain on wild fish supplies, aquaculture and the farming of aquatic organisms like fish, crustaceans, mollusks, and aquatic plants have become an essential part of the world's food production. The survival and development of aquatic creatures, however, depend critically on the maintenance of ideal water quality in aquaculture systems; deviations can cause stress, disease outbreaks, and decreased productivity.

The recent convergence of machine learning (ML) algorithms and Internet of Things (IoT) technology has revolutionized several sectors by providing creative solutions to challenging issues. The fusion of IoT sensors and machine learning techniques opens up a previously unheard-of possibility for aquaculture: real-time monitoring, analysis, and prediction of water quality parameters that will maximize operational effectiveness, reduce environmental impact, and guarantee the sustainability of aquaculture practices.

The importance of managing water quality in aquaculture is examined in this research, along with the difficulties in using conventional monitoring techniques and the promise of IoT-enabled sensors in conjunction with ML algorithms to transform water quality prediction. ML algorithms can identify intricate patterns and correlations by utilizing the enormous data gathered by IoT sensors placed within aquaculture facilities. This allows for the precise forecasting of important water quality indicators including temperature, dissolved oxygen, pH, ammonia levels, and turbidity.

Additionally, the predictive powers provided by machine learning algorithms enable aquaculture managers to take proactive steps to reduce variability in water quality, optimize feeding schedules, reduce the risk of disease, and improve overall productivity. Additionally, ML models can adjust to changing environmental conditions by using past data and continuously learning from fresh observations. This allows them to provide actionable insights for well-informed decision-making in real time.

2. LITERATURE REVIEW

Several studies have suggested using machine learning techniques for this specific function. One such study is that of Yang et al., who presented a hybrid prediction model that combines a deep belief network (DBN) and variational mode decomposition (VMD). The DBN and VMD were further refined using an enhanced slime mold method [1]. Additionally, a study by V.P.et al. proposed a smart aquaculture system that predicts water quality using k-means clustering and a deep convolutional neural network (D-CNN) [2]. Using LSTM (Long-Short Term Memory) method improves the prediction of aquaculture water quality. Temporal dependencies is emphasized by the LSTM-GRU (Gated Recurrent Unit) combination [3]. In paper [5], the authors offer a novel LSTM method that improves pH and water temperature forecasts by using pre-processed data and correlation insights. In,[4] the authors develop an intelligent aquaculture water quality inspection system using AIoT (Artificial Intelligence of Things). By employing a deep learning model for predicting the water quality environment to anticipate the data, it addresses the sensor drift brought on by the increasing number of seawater algae. The experiment on internal data indicated a positive impact on the application of artificial intelligence in aquaculture, helping to cut costs and time and boost farming efficiency. An end-to-end deep learning based TabNet (Tabular Network) model for water quality prediction was developed [6]. The stacked ensemble H2O AutoML model is used in an automated water quality prediction system that effectively handles missing variables and produces water quality predictions with good accuracy [7]. In paper [8] the author explores the possible applications of IoT in aquaculture and looks at how it can improve disease prevention and water quality monitoring. It emphasizes the revolutionary potential of IoT technology in providing real-time data on water parameters and enabling preventative measures against diseases. The authors have suggested a pathplanning algorithm to sample every pond on the property while using the fewest possible resources (i.e., the quantity of UAVs and their power consumption)[9]. After examining the several real-world sensor datasets gathered from the Internet of Things were used to assess four deep learning anomaly detection techniques: autoencoder, variational autoencoders, long short-term memory autoencoder, and spectral-residual convolutional neural network [11]. We also can see that the sensors can monitor the water quality in a fish farm using mobile devices thanks to wireless transmission technology that uses a variety of sensors to transfer data to the server, including temperature, pH value, dissolved oxygen, water level, and life expectancy of the sensor [10]. A buoy system with artificial intelligence (AI) that gathers pertinent water quality data on its own, sends it right away to the shore server over wireless channels, and assists server-side AI algorithms in using machine learning techniques to provide more short-term water quality predictions.[12]. Based on percentage error (PE) and prediction accuracy (PA), the Autoregressive Integrated Moving Average's (ARIMA) predictive analytics are evaluated. We describe an Internet of Things (IoT)-based Biofloc monitoring system that can do predictive and stream analytics at a cheaper cost [13]. The use of multivariate linear regression, random forests, and artificial neural networks to analyse data from water quality indicators, such as pH, dissolved oxygen, and pond temperature, that are regularly recorded in fish farming. Machine learning (ML) methods have been created to forecast the dynamics of such elements, thereby assisting fish farmers in making more informed decisions [14].

3. MATERIALS AND METHODOLOGY

Our approach to aquaculture water quality prediction uses ML and IoT Technology to generate a dependable predictive model and real time monitoring system. To the mean of the two steps the major role of computation of data was also the main part. Thus, the project was mainly divided into 3 phases:

Phase 1: IoT Approach
Phase 2: Data Computation
Phase 3: Prediction

Phase 1:

3.1 Hardware Components

The smooth operation of our project depends on the flawless integration of sensors, microcontrollers, and other smart devices. The following hardware components must be used effectively for our project to function successfully:

- **Arduino Uno**: Acting as the main microcontroller for the project, the Arduino Uno makes it easier to organize and handle a variety of duties. It is the foundation of our system because of its adaptability and dependability, which provides effective control and communication between all connected components.
- **pH sensor**: This vital instrument measures the environment's acidity or alkalinity in real time. We can keep an eye on water quality, and other pertinent factors that are essential to achieving the goals of our project by precisely measuring pH levels. Its accuracy guarantees accurate facts for in-depth examination and judgment.
- **DS18B20 Temperature Sensor**: This temperature sensor is essential to our research because it is specifically made to detect temperature variations with a high degree of accuracy. This sensor offers accurate data necessary for thorough analysis and efficient decision-making procedures, regardless of whether it is tracking system temperature fluctuations or ambient temperatures.
- **Turbidity Sensor with Module**: This sensor module finds suspended particles or other contaminants in fluids to determine their clarity. It is very useful for real-time fluid condition analysis and water quality analysis due to its precise turbidity measurement capability. This sensor adds crucial information for well-informed decision-making about environmental monitoring or fluid management by offering insights into the degree of turbidity.
- **I2C LCD Module**: This module is like a small screen that shows us information. It's useful because it lets us see data from the sensors in real-time, like a mini dashboard. We can easily read important information without having to connect to a computer or other device.

3.2 Hardware Connection

We purchased required hardware, tools and connected them according to the shown in Fig. 3.2.1

Then we write the code in Arduino IDE (Integrated Development Environment) according to our hardware connection and upload it to the circuit board.

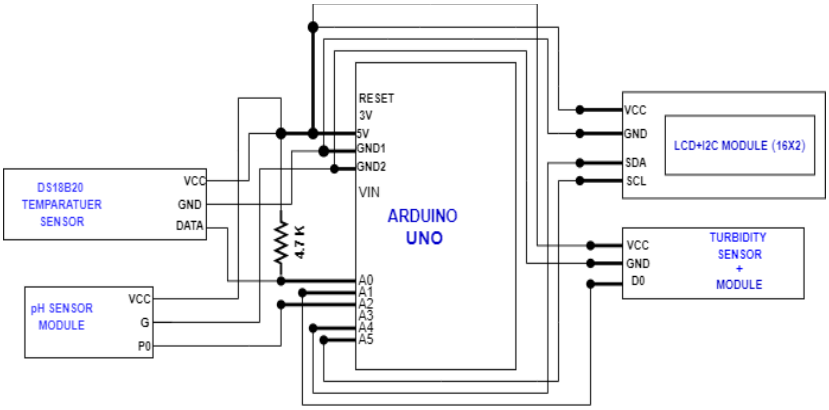


Fig 3.2.1 : Circuit Connection Setup

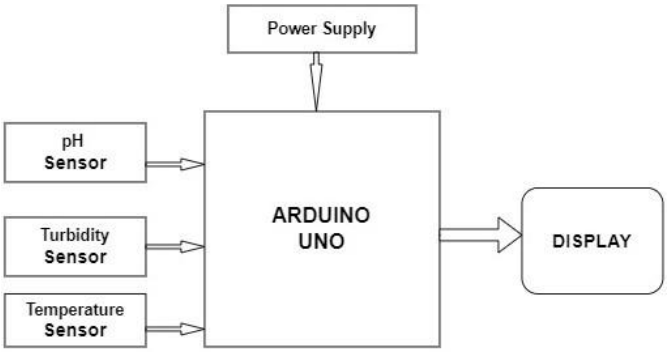


Fig 3.2.2 : Circuit Diagram

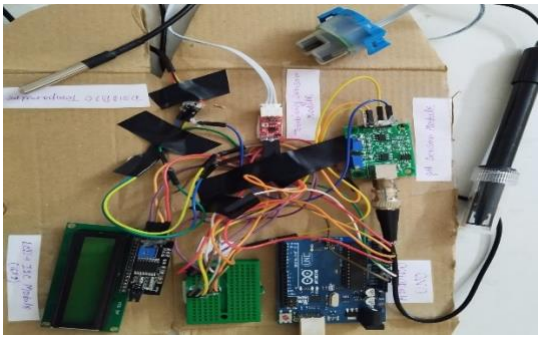


Fig 3.2.3: Hard Ware Connection

3.3 Model Architecture

The smooth operation of our project depends on the flawless integration of sensors, microcontrollers, and other smart devices. The following hardware components must be used effectively for our project to function successfully:

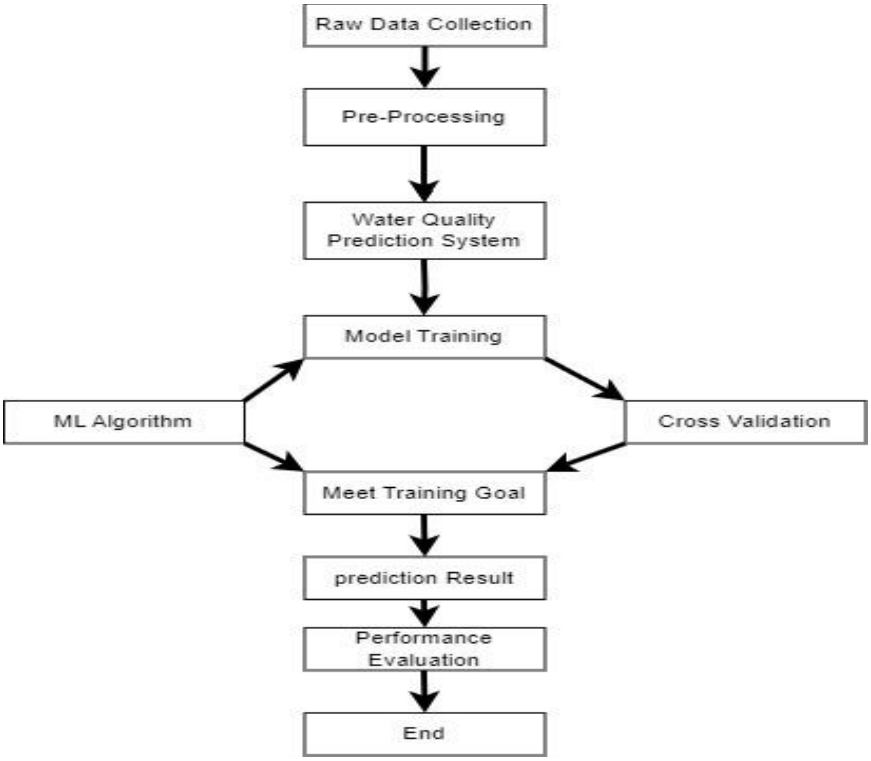


Fig 3.3.1: Model Architecture Flow Chart

Phase II

3.4 Study area

As the project was done in short term, so we could only collect water from all the ponds and fisheries near by our campus that is Adamas University, almost 100 water samples were collected. Fig 3.4.1 all are the glimpse of the waterbodies where water was collected for the experiment.



Fig 3.4.1(A) Snapshot various of water bodies

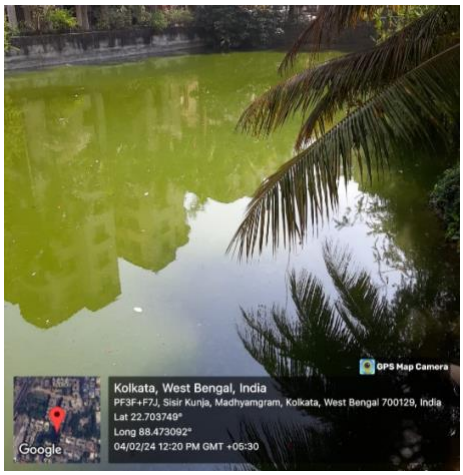


Fig 3.4.1(B) Snapshot of various water bodies



Fig 3.4.1(C) Snapshot of various water bodies

3.5 Data Collection

We collected samples of water from different ponds near our campus. We bought instruments and sensors for data collection. We used pH, waterproof temperature, and Turbidity sensors to measure the values. Sample collection was more manageable but the hardware connection and the data preprocessing were difficult and time-consuming we worked for more than two months to process the necessary data from the samples.

The computation of Dissolved Oxygen was the most time-consuming as the sensors are quite costly, so we went for the experimental lab approach. the following steps and requirements are mentioned below:

Requirements

- Water sample
- BOD free water
- Allylthiourea solution (0.5%)
- Phosphate buffer solution (pH 7.2)
- Sulfuric acid (1N) (2.8 ml of conc. H_2SO_4 added to 100 ml of BOD free distilled water)
- Sodium hydroxide (1N) (4g of NaOH dissolved in 100 ml of distilled water)
- Reagent for DO estimation
- BOD bottles (6)
- Flask
- Pipette
- BOD incubator
- pH meter

Procedure

- Adjust the pH of the water sample to neutrality using 1N acid or 1N alkali solutions.
- Fill the water sample in 6 BOD bottles without bubbling.
- Add 1 ml of allylthiourea solution to each bottle.
- Determine dissolved oxygen content in three of the 6 BOD bottles by the titration method.
- Collect water samples without bubbling in the 250 ml glass bottle.
- Add 2 ml each of manganous sulfate and alkaline iodide-azide solutions in succession, right at the bottom of the bottle with separate pipettes, and replace the stopper.
- Shake the bottle in the upside-down direction at least six times.

- Allow the brown precipitates to settle.
- Add 2 ml of concentrated sulfuric acid and shake the stoppered bottle to dissolve the brown precipitates.
- Take a 50 ml sample in a flask and titrate it with thiosulfate solution till the color changes to pale to blue.
- Add 2 drops of starch solution to the above flask which changes the colour of the contents from pale to blue.
- Titrate again with thiosulfate solution till the blue color disappears.
- Find out the total amount of sodium thiosulfate solution (titrant) used and calculate the dissolved oxygen content of water (mg/L) by applying the equation:

D.O. (mg/L) =((8×1000×N)/V)

Where V = volume of Water sample taken (ml)

N = normality of the titrant (0.025) [titrant = Sodium thiosulphate]

8 = the constant since 1 ml of 0.025 sodium thiosulfate solution is equivalent to 0.2 mg oxygen

Table 3.5.1: Titration reading for sample No.1

Table For Sample No. 1					
No. of Observations	Volume of Sampel(ml)	Burette reading			Mean (ml)
		Initial Value	Final Value	Difference	
1.	50	0	1.5	1.5	8.5/5 = 1.7
2.	50	1.5	3.5	2.0	
3.	50	3.5	5.5	2.0	
4.	50	5.5	7.2	1.7	
5.	50	7.2	8.5	1.3	

Calculation:
Dissolved Oxygen (mg/l) = (1.7 x 0.025 x8 x1000) / 50 = 6.8 (~6.766465)

Table 3.5.2: Titration reading for sample No.2

Table For Sample No. 2					
No. of Observations	Volume of Sampel(ml)	Burette reading			Mean (ml)
		Initial Value	Final Value	Difference	
1.	50	0	1.5	1.5	9.2/5 = 1.84
2.	50	1.5	3.9	2.4	
3.	50	3.5	5.9	2.0	
4.	50	5.5	7.7	1.8	
5.	50	7.7	9.2	1.5	

Calculation:
Dissolved Oxygen (mg/l) = (1.84 x 0.025 x8 x1000) / 50 = 7.36 (~7.365426)



Fig 3.5.1 Solution for titration

The figure 3.5.1 suggests that the solution was prepared for titration. This was done in the Lab of School Of Life Science, Adamas University Under the guidance of lab experts. The value that we got was approximately 6.46 ml/l where the Ph -temp- and turbidity – were the assigned value.

To find all the Dissolved Oxygen value experimentally was time taking job so with the help of empirical formulation:

DO = a * np. power (pH, b) * np.exp(c * T) * d * np.power(Turbidity, e) … eqn3.1

Table 3.5.3 describes the values of the equation 3.1

Sl No.	Terms	Description
1.	pH	pH value
2.	T	Temperature
3.	Turbidity	Turbidity of the value
4.	a, b, c, d, e	Co-efficient

The relation is established between all the values in table 3.5.3 as:

Terms related to pH and Temperature: "a * np. Power (pH, b) * np. exp (c * T)" implies that pH and temperature have an impact on dissolved oxygen. The exponential relationship between pH and dissolved oxygen is represented by the np. Power () function, which shows that variations in pH have a nonlinear impact on dissolved oxygen concentrations. The exponential relationship between temperature and dissolved oxygen is represented using the np. Exp () function, which shows that dissolved oxygen tends to drop exponentially with increasing temperature.

Turbidity Term: It appears from the term "d * np. Power (Turbidity, e)" that turbidity also affects dissolved oxygen. This phrase suggests that variations in turbidity may have an impact on dissolved oxygen levels; the nature of the impact may be nonlinear, contingent upon the coefficient 'e' value.

The co-efficient a, b, c, d, and e were taken from the values of Ph, Temperature and turbidity of the data collected with the help of sensors. We have taken the values and found the co-efficient variation and taken the round-off values of it.

```
import statistics

# Define your values
values = [7.32, 5.33, 6.49, 7.36, 7.64]

# Calculate the coefficient of variation
coefficient_of_variation = (statistics.stdev(values) / statistics.mean(values))

print("Coefficient of Variation:", coefficient_of_variation)
```

```
Coefficient of Variation: 0.13782958695394093
```

```
# Original number
number = 0.137

# Round to two decimal places
rounded_number = round(number, 2)

# Print the rounded number
print(rounded_number)
```

```
0.14
```

fig 3.5.2 Snapshot of co-efficient and the round-off value.

To predict water quality of the sample the following equation is used which is again an empirical formulation on the bases of theoretical relation. The equation as follows:

Water Quality = [(0.2*Temp) +(0.1*pH) + (0.15*Dissolved Oxygen) - (0.05*Turbidity)] …eqn 3.2

Table 3.5.4: describes the equation 3.2.

Sl No.	Variables	Co-efficient	Effect on Water Quality
1.	Temperature	0.2	0.2°C Increase in Temp, water Quality increases by 0.2 unit
2.	pH	0.1	0.1 Increase in pH, water Quality increases by 0.1 unit
3.	Dissolved Oxygen	0.15	Increase in 1 unit of DO, 0.15 unit increase in water quality
4.	Turbidity	0.05	Increase of 1 unit of turbidity, 0.05 unit drop in water quality

3.6 Data Preprocessing

The smooth operation of our project depends on the flawless integration of sensors, microcontrollers, and other smart devices. The following hardware components must be used effectively for our project to function successfully describe the code and the empirical solution

3.6.1 Dependencies (libraries)

Python was the core language for handling the dataset to visualize and predict the values for the desired model. The libraries that were used for gathering the dataset to predict were:

a) Statistics

b) Pandas

c) NumPy

d) matplotlib

e) seaborn

3.6.2 Parameters &Algorithm

Table 3.6.2.1: describes all the parameters used in the project work.

Feature	**Data Type**	**Potential Values**	**Description**
Sample no.	Int	Unique Identifier	ID no. of the water sample.
pH	Float	Any integer value	pH value of each sample was taken 5 times a day
Average Ph	Float	Any integer value	The average pH value of each sample.
Temperature	Float	Any integer value	Temperature value of each sample was taken 5 times a day
Average Temp	Float	Any integer value	The average Temperature value of each sample.
Turbidity (NTU)	Float	Any integer value	0-20- muddy 20-40-misty 40-60-clear
Clearance	Character	1,2,3	1-muddy, 2-misty, 3-clear
Dissolved Oxygen	Float	Any integer value	0-5, 10 and above not fit, 5-10- fit for survival
Water Quality	Float	Any integer value	Need to be predicted

Algorithm:

Linear regression is straightforward, easy to understand, and highly effective, it is frequently used for aquaculture prediction. The input factors (water temperature, pH levels, turbidity) and output variables (dissolved oxygen, survival rates) in aquaculture data frequently show linear connections. Making decisions is facilitated by linear regression, which makes it simple to understand how each input variable affects the result. Furthermore, it can be scaled to big datasets that are frequently seen in aquaculture due to its computing efficiency. Although it might not be able to capture intricate relationships, its simplicity in usage and comprehension renders it a useful instrument for aquaculture predictive modelling.

4. RESULT ANALYSIS

A topic analysis like "water quality prediction using IoT and ML" would normally include a number of essential elements meant to assess the efficacy and performance of the predictive model created with machine learning methods and information gathered from Internet of Things (IoT) sensors. An overview of a typical result analysis procedure is provided below:

4.1 MODEL PERFORMANCE EVALUATION:

The objective of this investigation was to use machine learning models to forecast water quality based on a variety of environmental parameters, including temperature, pH, dissolved oxygen, and turbidity. The dataset was loaded from an Excel file for analysis, and the data was preprocessed, machine learning models were trained, and their effectiveness was assessed.
Data Loading and Preprocessing: The Excel file included the dataset that was loaded.
Environmental parameters (temperature, pH, dissolved oxygen, turbidity) and the accompanying water quality labels make up the dataset.
As part of the data preprocessing step, the dataset was divided into features (X) and the target variable (y), and then an 80-20 split was used to separate the training and testing sets.
Model Development and Assessment:

	Sample No.	PH	(°C)	(NTU)	Dissolved_Oxygen	WaterQuality
0	1	7.32	28.63	50.00	6.833203	4.982980
1	2	5.33	29.07	48.00	7.435105	5.062266
2	3	6.49	28.65	55.00	6.858873	4.657831
3	4	7.36	29.16	30.45	7.560132	6.179520
4	5	7.64	29.16	42.34	7.587927	5.617189
..	...	...	...	...	...	...
95	96	7.36	28.63	41.00	6.820027	5.435004
96	97	7.49	28.68	39.34	6.886929	5.551039
97	98	7.21	29.16	30.45	7.558575	6.164286
98	99	5.64	29.07	48.00	7.439309	5.093896
99	100	7.71	28.65	55.00	6.870698	4.781605

Fig 4.1.1: All Collected Data

```
Missing Values:
Sample No.          0
      PH            0
 (°C)               0
 (NTU)              0
 Dissolved Oxygen   0
Water Quality       0
dtype: int64
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Sample No.         100 non-null    int64
 1         PH           100 non-null    float64
 2    (°C)              100 non-null    float64
 3    (NTU)             100 non-null    float64
 4    Dissolved Oxygen  100 non-null    float64
 5   Water Quality      100 non-null    float64
dtypes: float64(5), int64(1)
memory usage: 4.8 KB
Number of Duplicate Rows: 0
Summary Statistics:
       Sample No.          PH        (°C)       (NTU)   Dissolved Oxygen
count  100.000000  100.000000  100.000000  100.000000         100.000000
mean    50.500000    7.185500   28.900200   42.866800           7.132333
std     29.011492    0.837334    0.251765    7.912998           0.354840
min      1.000000    5.330000   28.630000   30.450000           6.736469
25%     25.750000    7.320000   28.630000   39.340000           6.765828
50%     50.500000    7.425000   29.070000   42.340000           7.365488
75%     75.250000    7.640000   29.160000   48.000000           7.495063
max    100.000000    8.070000   29.160000   55.000000           7.519701
```

Fig 4.1.2: describes all the description and mising values of the variables used for prediction.

Regarding Linear Regression:
As a continuous variable, the model was trained to predict the quality of the water.
For regression tasks, Mean Squared Error (MSE) was used to evaluate the model.

Outcomes: Linear Regression: The testing data's Mean Squared Error (MSE) was roughly 0.168 and for training data was nearly 0.205.

```
Mean Squared Error (MSE) on training data: 0.201526715984942
Mean Squared Error (MSE) on testing data: 0.16792837214550088
```

Fig 4.1.3: Mean Squared Error

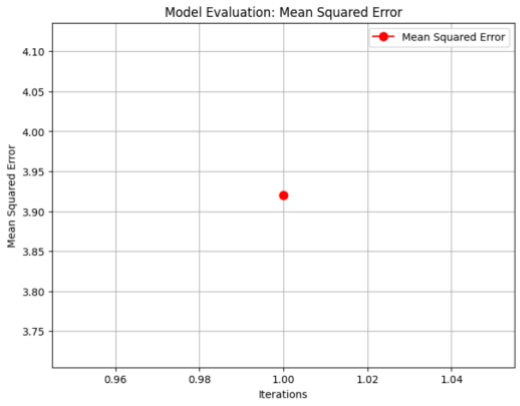


Fig 4.1.4: model evaluation

Analysis: Linear Regression: The MSE value derived from this technique indicates that, on average, the model's predictions differ by 0.168 units from the actual water quality measurements.

Investigating the distribution of errors and possible causes of higher errors could be part of a more thorough analysis.

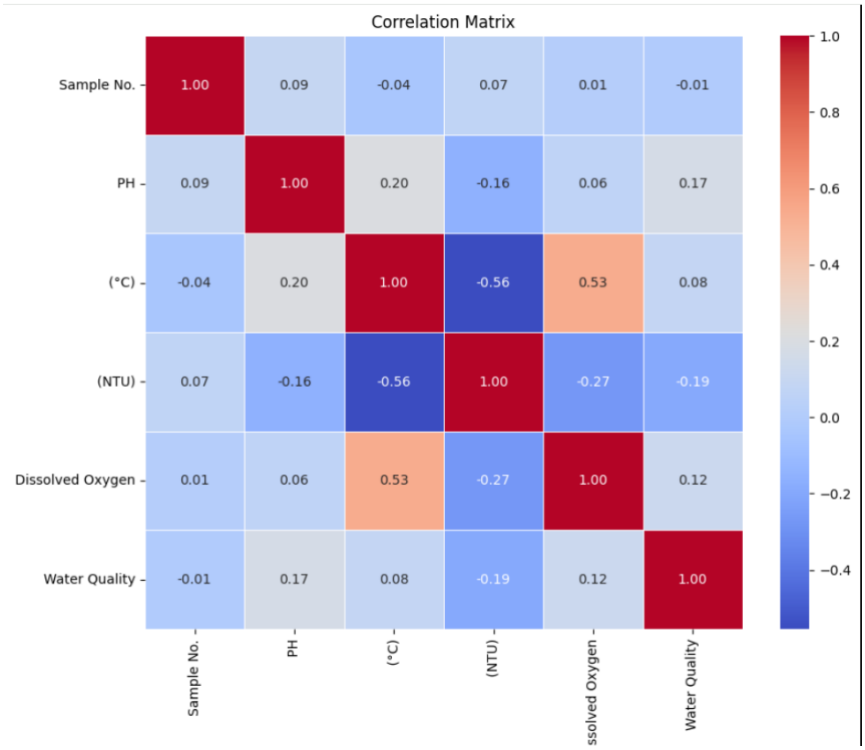


Fig 4.1.5: Corelation Matrix

In summary, even if the models produced encouraging results, more research and validation are required to guarantee the validity and applicability of the conclusions. Furthermore, investigating other assessment metrics and methods might offer more in-depth understanding of the model's functionality and possible areas for development.

```
Accuracy: 1.0
Classification Report:
              precision    recall  f1-score   support

           1       1.00      1.00      1.00        20

    accuracy                           1.00        20
   macro avg       1.00      1.00      1.00        20
weighted avg       1.00      1.00      1.00        20
```

Fig 4.1.6: Accuracy (F1–Score)

4.2 FEATURE IMPORTANCE ANALYSIS:

• Assess the significance of several features (such as pH, temperature, and turbidity) in forecasting the quality of the water. The most important variables can be found using methods like permutation importance or feature importance plots.
• Impact of feature engineering: Evaluate how feature engineering methods affect model performance. This covers transforming, scaling, or producing new features from unprocessed sensor input.

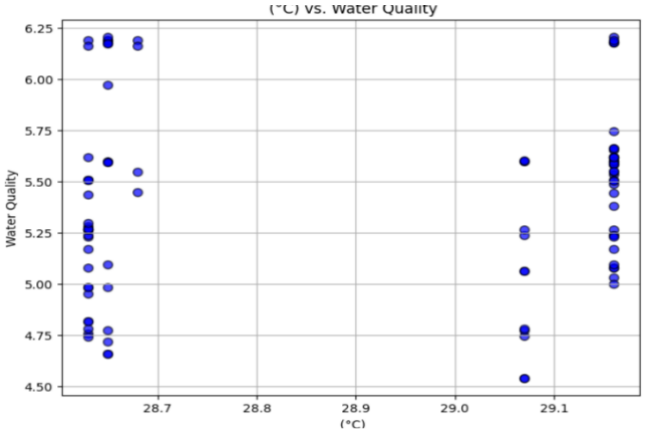


Fig 4.2.1: Plotting Water Temperature Vs. Water Quality

4.3 IoT sensor Performance:

• Sensor accuracy and reliability: Assess how well IoT sensors measure different aspects of water quality. Comparing sensor results with reference data from laboratory analysis or high-quality sensors may be necessary to achieve this.
• Data quality assessment: Look at the flaws, such as missing values, outliers, or sensor drift, and how they impact model predictions in the data gathered by IoT sensors.



Fig 4.3.1 IOT Sensor Performance

4.4 Visualization and Interpretation:

• Visualise predictions: To show how the model's predictions compare to real water quality

measurements, use visualisations like time series plots, scatter plots, or heatmaps.
• Interpretability: Learn how to read machine learning models to comprehend how they generate predictions, particularly for stakeholders who might not be familiar with the basic ideas behind machine-learning.

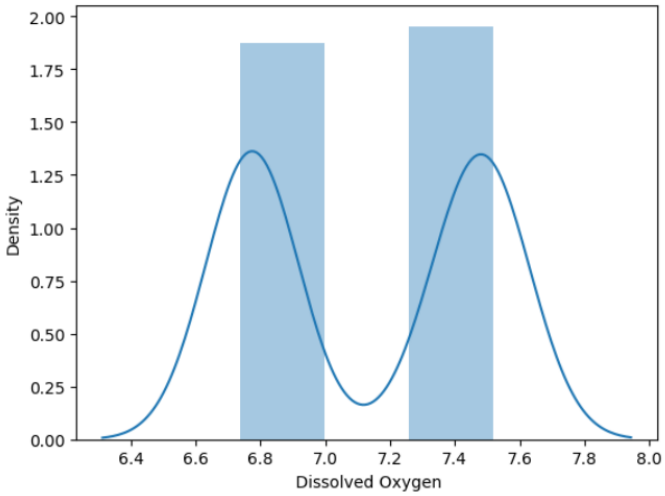


Fig 4.3.2: Dissolved Oxygen Density Diagram

Overall, a thorough assessment of the predictive model's performance as well as the efficiency of IoT sensors in gathering and supplying data for precise forecasts is part of the result analysis of water quality prediction combining IoT and ML. It seeks to shed light on the viability and possible consequences of implementing such systems in practical environmental monitoring applications.

5. FUTURE WORK

1. Considering real-world deployment:

- Scalability: Consider how effectively the IoT infrastructure and machine learning model scale for real-world deployment in larger areas or with more sensors.
- Cost-effectiveness: Compare the savings from using the recommended solution to traditional monitoring methods, accounting for costs related to sensor installation, maintenance, and data management.
- User input: Consult end users, such as environmental agencies or water management authorities, to gauge the system's feasibility and usefulness.

`2. Prospective Routes:

- Discuss the limitations: Discuss any limitations or challenges you encountered when carrying out the study, such as the interpretability of the models, sensor precision, or data accessibility.

- Research directions for the future: Make recommendations for future lines of inquiry that will improve the accuracy, robustness, and usefulness of machine learning- and IoT-based water quality prediction systems. These can involve incorporating environmental characteristics, integrating sensor fusion techniques, or researching complex machine learning algorithms.

6. CONCLUSION

In conclusion, the utilization of Water quality prediction in Aquaculture using ML and IoT (WQPA) signifies a transformative breakthrough in aquaculture management. A smooth tie up among the power of real-time data collection and analysis, these systems empower aqua culturists to maintain optimal water quality conditions crucial for the growth and health of aquatic species. Through proactive monitoring and precise adjustments, WQPA enhances productivity and sustainability in aquaculture operations, while also reducing the ecological impact. Moreover, the scalability and affordability of IoT technology make WQPA accessible to aquaculture facilities of all sizes. As we navigate the challenges of feeding a growing global population sustainably, WQPA offers a promising solution to improve efficiency, minimize risks, and promote responsible stewardship of aquatic resources. By embracing this technology, the aquaculture industry can cover the way for a strong and thriving future, ensuring the continued growth and development of this vital sector.

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